

# CariSurg MedTech Pathways Research Project

## Preliminary Proposal

---

**Project Title:** AI-Assisted Triage Support Tool for High-Risk Patient Identification in the Emergency Department

**Student Name:** Mithra Ramoutar

### Statement of the Problem

Mercer General Hospital's ED faces risk of patient under-triage during peak periods, where nurse workload and time pressure can cause high-risk cases to be missed. A 12-week pilot will test an explainable AI support tool that uses structured triage data and free-text presenting complaints to flag high-risk cases for nurse review, displaying a visible reason for each alert. The pilot will measure sensitivity, specificity, alert burden, and fairness across demographic groups, with nurses retaining full accountability for all final escalation decisions.

### Previous Work

A growing body of literature has examined the application of artificial intelligence to emergency department triage. The following six studies form the evidence base for this proposal.

#### Tyler et al. (2024)

Tyler et al. conducted a scoping review of AI use in hospital emergency department triage. The clinical problem addressed was the combination of overcrowding, variable triage decisions, and pressure to identify high-risk patients quickly. The review found that AI and machine-learning tools show promise for acuity prediction, resource allocation, and earlier recognition of patients requiring urgent care. However, the authors note that the evidence base remains early-stage, with most studies relying on retrospective data rather than live clinical testing. This limits conclusions about real-world effectiveness and safety.

#### Da'Costa et al. (2025)

Da'Costa et al. reviewed the benefits, challenges, and future directions of AI-driven triage. The paper demonstrates that AI could improve triage speed, consistency, and decision support, particularly in high-volume departments where clinicians must make rapid judgements with limited information. The authors identify concerns around bias, data quality, explainability, privacy, and clinician trust. A key finding is that strong technical performance does not automatically ensure that an AI tool is safe, fair, or practical within real emergency department workflows, a gap directly relevant to this proposal.

### **Araouchi & Adda (2024)**

Araouchi and Adda developed TriageIntelli, a multimodal AI triage system evaluated on 1,267 patient records using the Korean Triage and Acuity Scale (KTAS). They trained and compared six machine-learning models including, SVM, Random Forest, ANN, GBM, Logistic Regression, and XGBoost, finding that SVM and GBM achieved the highest individual accuracies of 78.7% and 79.0% respectively. A stacking model combining the top three performers reached 80.05% accuracy and an AUROC of 0.93. A key limitation noted by the authors is that the models were trained on a single dataset, which may reduce generalisability to other clinical environments with different demographics or care contexts.

### **Chen et al. (2024)**

Chen et al. developed and tested machine-learning and natural language processing models to predict patient disposition using both structured triage data and free-text clinical notes. The problem addressed was that useful prognostic information present in brief nurse-written notes is rarely exploited by existing triage tools. Their models outperformed emergency physicians and a logistic regression baseline for several disposition predictions. The authors note that their work focused mainly on immediate disposition outcomes and did not fully address longer-term outcomes, resource use, waiting time, or patient survival.

### **Ottenhoff et al. (2023)**

Ottenhoff et al. investigated whether machine learning could serve as an adjunct to standard emergency department triage. The clinical problem was that traditional triage systems may miss patients whose risk only becomes apparent when multiple early clinical signals are considered together. Their findings suggest that machine-learning models can help identify patients at risk of adverse outcomes and support more consistent triage decisions. A key limitation is that model performance alone does not indicate whether clinicians will trust, use correctly, or benefit from such tools in a live triage environment.

### **Townsend et al. (2023)**

Townsend et al. explored medical practitioner perspectives on AI in emergency triage. The study addressed not only whether AI can predict risk, but whether clinicians believe it can fit safely into urgent clinical decision-making. Practitioners saw potential value in AI support, especially for consistency and risk recognition, but raised concerns about accountability, transparency, bias, and over-reliance. This paper is particularly relevant because it demonstrates that implementation challenges are not purely technical, they also depend on clinician trust, workflow design, and the clear assignment of responsibility for final decisions.

## Gaps in the Literature

Two gaps in the existing literature support the case for this pilot:

- Many AI triage studies report promising retrospective model performance, but fewer test whether these tools improve live nurse triage decisions in real emergency department workflows. The proposed pilot addresses this by embedding the tool directly into the triage process and measuring its effect on escalation decisions.
- Several studies focus on accuracy metrics, but fewer evaluate alert burden, fairness across patient groups, and whether clinicians can safely understand and act on AI recommendations. This pilot addresses this by measuring explainability, demographic fairness, and alert volume alongside predictive performance.

## Proposed Solution

This pilot proposes the development and evaluation of an explainable AI support tool for use during emergency department triage at Mercer General Hospital. The tool will process structured triage inputs, including vital signs, chief complaints, and acuity scores alongside free-text presenting complaint notes to identify patients at elevated risk of under-triage.

Drawing on approaches demonstrated by Araouchi and Adda and Chen et al. , the model will combine gradient boosting and natural language processing to flag high-risk patients for nurse review. Each alert will display a visible, human-readable reason (e.g., elevated heart rate combined with complaint keywords associated with sepsis), ensuring that nurses can evaluate and override the recommendation with full clinical accountability.

The 12-week pilot will measure the following outcomes:

- Sensitivity and specificity of high-risk flagging compared to retrospective clinical outcomes
- Alert burden (number of flags per shift) to ensure the tool does not overwhelm triage nurses
- Fairness across demographic groups to detect and mitigate potential bias in predictions
- Effect on escalation decision time, assessed against baseline triage workflow data

## References

1. Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus*, 16(5), e59906. <https://doi.org/10.7759/cureus.59906>
2. Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics*, 197, 105838. <https://doi.org/10.1016/j.ijmedinf.2025.105838>
3. Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted multimodal triage system for health centers. *Procedia Computer Science*, 251, 430-437. <https://doi.org/10.1016/j.procs.2024.11.130>
4. Chen, C. Y., Wu, K. H., Hung, S. K., Hsieh, M. J., Huang, E. P. C., Hsiao, C. T., Chiang, W. C., & Wang, H. C. (2024). Using machine learning and natural language processing in triage for prediction of clinical disposition in the emergency department. *BMC Emergency Medicine*, 24, 202. <https://doi.org/10.1186/s12873-024-01152-1>
5. Ottenhoff, M. C., et al. (2023). Machine Learning as an Adjunct to Traditional Triage in the Emergency Department. *Journal of Emergency Medicine*, 64(1), 83-92. <https://doi.org/10.1016/j.jemermed.2022.10.002>
6. Townsend, B. A., et al. (2023). Medical practitioner perspectives on AI in emergency triage. *Frontiers in Digital Health*, 5, 1297073. <https://doi.org/10.3389/fdgth.2023.1297073>